

John Bergdall

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Employment

2025-	The University of Arkansas , Associate Professor
2022-25	The University of Arkansas , Assistant Professor
2018-22	Bryn Mawr College , Assistant Professor
2017-18	Michigan State University , Visiting Assistant Professor
2014-17	Boston University , NSF Postdoctoral Research Fellow
2013-14	Boston University , Postdoctoral Faculty Fellow

Education

2008-13	Brandeis University , Ph.D. Advisor: Joël Bellaïche
2003-08	University of Minnesota , B.S.

Other academic positions

2021-22	Max-Planck-Institut für Mathematik Visiting researcher (1 year)
2017	Max-Planck-Institut für Mathematik Visiting researcher (3 months) Institut des Hautes Études Scientifiques Visiting researcher (2 months)

National grants and fellowships

2024	National Science Foundation (DMS-2401152) “Conference: Modular forms, L -functions, and eigenvarieties”
2023-	National Science Foundation (DMS-2302284) “Collaborative Proposal: Slopes of modular forms and moduli stacks of Galois representations”
2020-25	Simons Foundation Travel Support for Mathematicians (MPS-TSM-00713782) “Eigenvarieties, automorphic forms, and Galois representations”
2014-17	National Sci. Foundation Math. Sci. Postdoctoral Res. Fellowship (DMS-1402005) “Aspects of the Langlands program via p -adic families of automorphic forms”

University awards and fellowships

2023-24	University of Arkansas Robert C. and Sandra Connor Endowed Faculty Fellowship
2022-23	University of Arkansas New Faculty Commendation for Teaching Commitment

16. [New phenomena arising from \$\mathcal{L}\$ -invariant of modular forms](#) (with Robert Pollack)
To appear in *Algebra and Number Theory*.
15. [A \$p\$ -adic adjoint \$L\$ -function and the ramification locus of Hilbert modular eigenvarieties](#) (with Baskar Balasubramanyam and Matteo Longo)
Tunisian J. Math. 7, 2025, no. 3-4, p. 515-588. DOI:[10.2140/tunis.2025.7.515](#)
14. [Huber rings and valuation spectra](#) in *Non-Archimedean Geometry and Eigenvarieties EMS Münster Lectures in Mathematics* 4, 2024, p. 1-53. DOI:[10.4171/mlm/4](#)
13. [On \$p\$ -adic \$L\$ -functions for Hilbert modular forms](#) (with David Hansen)
Mem. of the Amer. Math. Soc. 298, 2024, no. 1489, pp. iv+126. DOI:[10.1090/memo/1489](#)
12. [Reductions of 2-dimensional semi-stable representations with large \$\mathcal{L}\$ -invariant](#) (with Brandon Levin and Tong Liu)
J. Inst. Math. Jussieu 22, 2023, no. 6, p. 2619-2644. DOI:[10.1017/S1474748022000081](#)
11. [Slopes of modular forms and reducible Galois representations, an oversight in the ghost conjecture](#) (with Robert Pollack)
Proc. Amer. Math. Soc. Ser. B 9, 2022, p. 432-444. DOI:[10.1090/bproc/136](#)
10. [Reductions of some two-dimensional crystalline representations via Kisin modules](#) (with Brandon Levin)
Int. Math. Res. Not., 2022, no. 4, p. 3170-3197. DOI:[10.1093/imrn/rnaa240](#)
9. [Smoothness of definite unitary eigenvarieties at critical points](#)
J. reine angew. Math. (Crelle's J.) 759, 2020, p. 29-60. DOI:[10.1515/crelle-2017-0048](#)
8. [Upper bounds for constant slope \$p\$ -adic families of modular forms](#)
Selecta Math. 25, 2019, no. 4, Art. 59, pp. 24. DOI:[10.1007/s00029-019-0505-8](#)
7. [Slopes of modular forms and the ghost conjecture, II](#) (with Robert Pollack)
Trans. Amer. Math. Soc. 372, 2019, no. 1, p. 357-388. DOI:[10.1090/tran/7549](#)
6. [Slopes of modular forms and the ghost conjecture](#) (with Robert Pollack)
Int. Math. Res. Not. 2019, no. 4, p. 1125-1144. DOI:[10.1093/imrn/rnx141](#)
5. [An adjunction formula for the Emerton–Jacquet functor](#) (with Przemyslaw Chojecki)
Israel J. Math. 223, 2018, no. 1, p. 1-52. DOI:[10.1007/s11856-017-1611-y](#)
4. [A remark on non-integral \$p\$ -adic slopes for modular forms](#) (with Robert Pollack)
C. R. Math. Acad. Sci. Paris 355, 2017, no. 3, p. 260-262. DOI:[10.1016/j.crma.2017.01.012](#)
3. [Paraboline variation of \$p\$ -adic families of \$\(\varphi, \Gamma\)\$ -modules](#)
Compositio Math. 153, 2017, no. 1, p. 132-174. DOI:[10.1112/S0010437X16007831](#)
2. [Arithmetic properties of Fredholm series for \$p\$ -adic modular forms](#) (with Robert Pollack)
Proc. Lon. Math. Soc. 113, 2016, no. 3, p. 419–444. DOI:[10.1112/plms/pdw031](#)
1. [Ordinary modular forms and companion points on the eigencurve](#)
J. Number Theory 134, 2014, p. 226-239. DOI:[10.1016/j.jnt.2013.07.014](#)

Preprints

2. On the variation of slopes over moduli of Galois representations (with Robert Pollack).
1. On the distribution of modular points on deformation spaces (with Chengyang Bao and Brandon Levin).

Computational research code

- 2023- **Github repository:** [\$\mathcal{L}\$ -invariants of modular forms](#) (joint with Robert Pollack)
- 2021- **Github repository:** [Slopes of modular forms](#) (joint with Robert Pollack)
- 2017- **Website:** [Slopes of modular forms and Fredholm series](#) (joint with Robert Pollack)

Invited lecture series

- 2023 **Spring school on non-Archimedean geometry and eigenvarieties** (Heidelberg, DE)
Four graduate-level lectures on adic spaces
- 2022 **Graduate school on p -adic L -functions and eigenvarieties** (South Bend, IN)
Four graduate-level lectures on Galois representations and eigenvarieties

Conference and workshop presentations

- 2026 **Algebraische Zahlentheorie** (Oberwolfach, DE)
“On the distribution of modular points on deformation rings”
- 2023 **TORA XII: Tex.–Okla. Representations and Automorphic Forms** (Norman, OK)
“ p -adic distributions of modular forms”
- 2022 **PCMI: Number theory informed by computation** (Park City, UT)
“ p -adic distributions of modular forms”
 p -adic L -functions and eigenvarieties (South Bend, IN)
“ p -adic distributions of modular forms”
- 2020 **AMS special session on automorphic forms and Galois representations** (Virtual)
“Reductions of local Galois representations for eigenforms with large $\mathcal{L}$ -invariants”
- 2019 **Moduli spaces and modularity** (Oaxaca, MX)
“Explicit $\mathfrak{S}$ -modules for crystalline representations”
AMS special session on special values of L -functions and arithmetic invariants in families (Hartford, CT)
“Constant slope families of p -adic modular forms”
33rd Automorphic Forms Workshop (Pittsburgh, PA)
“Constant slope families of p -adic modular forms”
- 2017 **AMS special session on p -aspects of arithmetic geometry** (Buffalo, NY)
“Upper bounds for constant slope p -adic families of modular forms”
 p -adic methods for Galois representations and modular forms (Barcelona, ES)
“Geometric properties of p -adic families of automorphic forms (and applications)”

Conference and workshop presentations, cont.

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| 2016 | AMS special session on p-adic analysis in number theory (Minneapolis, MN)
“Some questions about slopes of modular forms”
Connecticut summer school in number theory (Storrs, CT)
“Geometric properties of p -adic families of automorphic forms and applications”
The p-adic Langlands program and related topics (Bloomington, IN)
“Slopes of modular forms and the ghost series” |
| 2015 | Boston University/Keio University joint workshop in number theory (Boston, MA)
“Slopes of modular forms and the ghost conjecture”
p-adic methods in the theory of classical automorphic forms (Montréal, CA)
“Arithmetic properties of Fredholm series” |
| 2014 | Fourth annual upstate New York number theory conference (Buffalo, NY)
“Ordinary representations on $U(3)$ and a conjecture of Breuil and Herzig” |
| 2013 | Modular forms, p-adic L-functions and Selmer groups (Oriahovitz, BG)
“Parabolizations over families of trianguline representations” |

Recent colloquia & seminar presentations († indicates colloquium or other special presentation)

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| 2026 | Inst. for Adv. Study
University of Arizona | “On the distribution of modular points on deformation rings”
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| 2025 | University of Chicago
Michigan St. University
Ohio State University
University of Arkansas | —
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— |
| 2024 | Duke University
Oklahoma University
Oklahoma University† | “Slopes and $\mathcal{L}$ -invariants of modular forms”
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“Some distributions arising in algebra and number theory” |
| 2023 | Rice University
University of Arkansas
University of Pittsburgh†
University of Arkansas | “ p -adic slope distributions of modular forms”
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“Huber rings and valuation theory” (3 lectures) |
| 2022 | Université de Lille
Univ. du Luxembourg
Max Planck Inst.
Universität Heidelberg
Indian Inst. Sci. (IISc)
Université Laval | “Recent investigations of $\mathcal{L}$ -invariants of modular forms.”
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| 2021 | Max Planck Inst.†
Boston Univ | “Problems in the non-Archimedean theory of modular forms”
“Reductions of certain semi-stable Galois representations” |
| 2020 | University of Oregon | “Reductions of some crystalline representations” |

Conference and workshop organization

- 2024 **Co-organizer:** [Modular forms, \$L\$ -functions, and eigenvarieties](#) (Paris, France)
2015 **Co-organizer:** [Boston University/Keio University workshop 2015](#) (Boston, MA)

Colloquia and seminars organized

- 2025- **Co-organizer:** University of Arkansas Algebra Seminar
 Co-organizer: University of Arkansas Teaching and Learning seminar
2025 **Organizer:** University of Arkansas Undergraduate Math Research Symposium
2022- **Organizer:** University of Arkansas graduate learning seminars (3 semesters)
2019-20 **Co-organizer:** [Bi-College math colloquium](#)
2019-20 **Co-organizer:** [Philadelphia area number theory seminar](#)
2013-17 **Co-organizer:** [Boston University number theory seminar](#)
2014-15 **Organizer:** Boston University graduate student learning seminars
 Topics: [The local Langlands conjectures](#), [\$p\$ -adic Hodge theory](#)

Recent Teaching

- 2022- **MATH 6190V:** Topics in algebra — elliptic curves
(UArk) **MATH 3-41103:** Introduction to Abstract Algebra I, II
 MATH 30803: Linear Algebra
 MATH 32003: Number Theory

2018-20 **MATH B102:** Calculus II
(BMC) **MATH B290:** Elementary number theory
 MATH B303-4: Algebra I, II
 MATH B317: Topics in advanced mathematics: elliptic curves
 MATH B398: Senior conference: mathematics and democracy
 MATH B399: Senior conference: mathematical cryptography
 MATH B503-4: Graduate algebra I, II

Degree bearing student advising and research

- 2026 **Luke Biddle** (AB, University of Arkansas Honors College '26)
 Thesis: *Primitive Pythagorean Triples in Lean and Reduction Modulo Odd Prime Powers*
2021 **Sandy Chen** (AB, Bryn Mawr College '21)
 Thesis: *The distribution of greatest common divisor of $\mathbb{Z}[\sqrt{2}]$*
2021 **Elsa Magness** (MA, Bryn Mawr College '21)
 Thesis: *An Exposition of the Sato–Tate Conjecture for Elliptic Curves with Complex Multiplication*
2020 **Sophia Schein** (AB/MA, Bryn Mawr College '20)
 Thesis: *Hecke operators on linear representations over finite fields*
2016 **Alexander Peraire-Bueno** (Boston University Academy thesis)
 Thesis: *Counting with partitions*

Other student research, mentorship, and service

- 2025- **Ph.D. Thesis Committees (University of Arkansas)**
Total number of students: 1.
- 2024- **Honors College Thesis Committees (University of Arkansas)**
Total number of students: 9.
- 2019 **Summer Science Research (Bryn Mawr College)**
Students: Sandy Chen and Sophia Schein
PROMYS research project writer
Students: Eric Tang, Aryaman Srikant, Emily Huang, and Aidan Griffin
Title: *Representation theory and Dickson's theorem*
- 2016 **PROMYS research project mentor**
Students: David Amirault, Vanshika Jain, Roshan Padaki, and Sabir Shaik
Title: *Slopes of Newton polygons*

College and university service

- 2026- **Promotion and Tenure Committee** (Univ. Ark. Math.)
- 2025 **Fulbright Program Campus Committee Member** (Univ. Ark.)
- 2023- **Curriculum committee** (Univ. Ark. Math.)
- 2023-25 **Ad-hoc graduate qualifying exam committee** (Univ. Ark. Math.)
- 2020 **Graduate Council** (Bryn Mawr Coll.)
Director of Graduate Studies in Mathematics (Bryn Mawr Coll., Math.)
- 2019-20 **Committee on Undergraduate Awards and Fellowships** (Bryn Mawr Coll.)

Non-college and -university professional service

- 2024- **Peer review for grants and other proposals**
National Science Foundation, Banff International Research Station
- 2016- **Peer review of publications**
Journals: 15 total including Algebra and Number Theory, Duke Math. J. Invent. Math.
- 2023 **TORA XII Graduate Student Q & A session**
Facilitated question and answer session for graduate student conference participants
- 2020 **Panel: Cross Atlantic representation theory and other topics online**
Topic: "How individuals are dealing with the pandemic"
- 2019 **Panel: Philadelphia Undergraduate Mathematics Conference**
Topic: Professional development
- 2016 **Panel: Boston College mathematics graduate student association**
Topic: Professional development

Substantial professional development

- 2024 **Inspiring Inquiry and Preparing Lifelong Learners** (Certificate provided by the Association of College and University Educators (ACUE))
- 2020 **Teaching and Learning Institute partnership**
Partners: Sara Grossman, Kirtee Ramo, and Yeipyeng Kwa
Online Teaching Institute (Bryn Mawr College)
- 2019 **Cornell Interactive Theater Ensemble workshop**
Theme: "Hang in There and Be Tough"
Posse Plus Retreat
Theme: "The State of our Union"
- 2018-19 **Teaching and Learning Institute partnership**
Partner: Jake Ogata Bernstein

Professional membership

- 2020-23 **Mathematical Association of America**
- 2008-24 **American Mathematical Society** (gap 2014-15)